Electromagnetic Induction

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Magnetic Flux

Magnetic Flux

Magnetic flux is proportional to number of magnetic field lines crossing a surface.

For uniform field, it is given by

$$\boxed{\phi = \vec{B} \cdot \vec{A}}$$

If there are 'n' number of turns, then

$$\boxed{\phi_{net} = n(\vec{B} \cdot \vec{A})} \qquad (\vec{B} = \text{uniform})$$

If field is not uniform, then

$$\boxed{\phi = \int \vec{B} \cdot \overrightarrow{dA}}$$

SI unit $\Rightarrow$ Tm^2 = weber (wb)

CGS unit $\Rightarrow$ Gauss cm^2 = maxwell

$1T = 10^4$ G

$1m^2 = 10^4\ cm^2$

1wb = 10^8 maxwell

Gauss Law for Magnetic Field

Net magnetic flux linked with any closed imaginary surface is always zero

$$\boxed{\oint \vec{B} \cdot \overrightarrow{dA} = 0}$$

Reason : Magnetic lines always form closed loops or monopoles do not exist.

Laws of Electromagnetic Induction

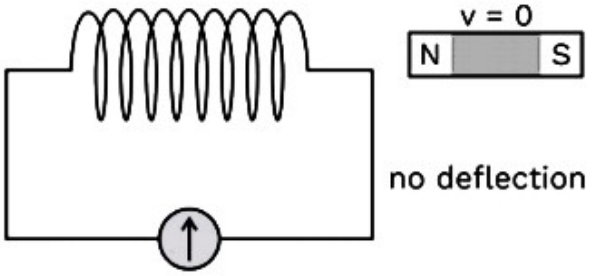
Electromagnetic Induction

The reverse effect of Oersted experiment was described by Michael Farady. He showed that when magnetic flux linked with a conductor changes, then an emf is induced across its ends. This phenomenon is called electromagnetic induction. This discovery brought revolution in field of generation of electric power.

Faraday's Experiment

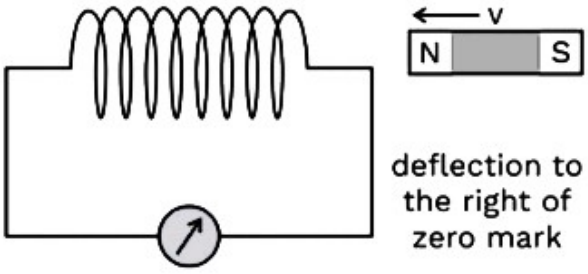
Faraday performed various experiments to discover and understand the phenomenon of electromagnetic induction. Some of them are :

(1) When the magnet is held stationary anywhere near or inside the coil, the galvanometer does not show any deflection.



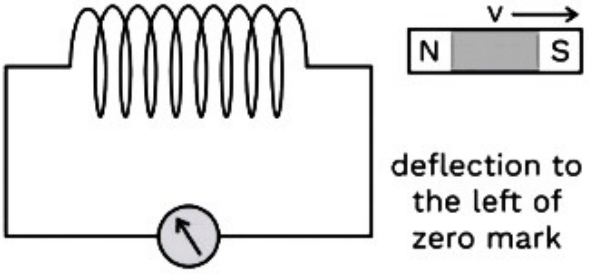
no deflection

(2) When the N-pole of a strong bar magnet is moved towards the coil, the galvanometer shows a deflection right to the zero mark.



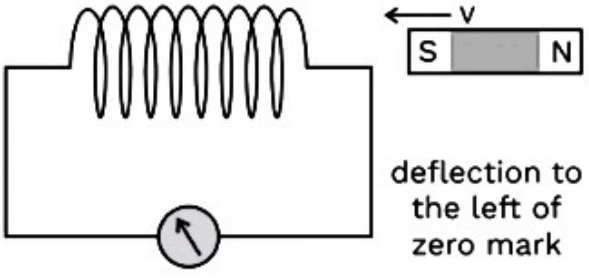
deflection to the right of zero mark

(3) When the N-pole of a strong bar magnet is moved away from the coil, the galvanometer shows a deflection left to the zero mark.



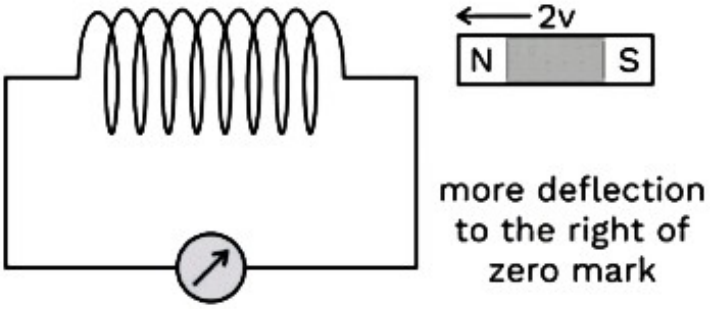
deflection to the left of zero mark

(4) If the above experiments are repeated by bringing the S-pole of the, magnet towards or away from the coil, the direction of current in the coil is opposite to that obtained in the case of N-pole.



deflection to the left of zero mark

(5) The deflection in galvanometer is more when the magnet moves faster and less when the magnet moves slow.



Whenever there is a relative motion between the source of magnetic field (magnet) and the coil, an emf is induced in the coil. When the magnet and coil move towards each other then the flux linked with the coil increases and emf is induced. When the magnet and coil move away from each other the magnetic flux linked with the coil decreases, again an emf is induced. This emf lasts so long the flux is changing.

Due to this emf an electric current start to flow and the galvanometer shows deflection.

The deflection in galvanometer last as long the relative motion between the magnet and coil continues.

Whenever relative motion between coil and magnet takes place an induced emf is produced in coil. If coil is in closed circuit then current and charge is also induced in the circuit. This phenomenon is called electromagnetic induction.

Faraday's Law of Electromagnetic Induction

Faraday observed that whenever magnetic flux related with coil changes with time some emf is induced in coil. The magnitude of emf is equal to magnitude of rate of change of magnetic flux with respect to time

$$|\varepsilon| = \left|\frac{d\phi}{dt}\right|$$

$$\varepsilon = -\frac{d\phi}{dt}$$

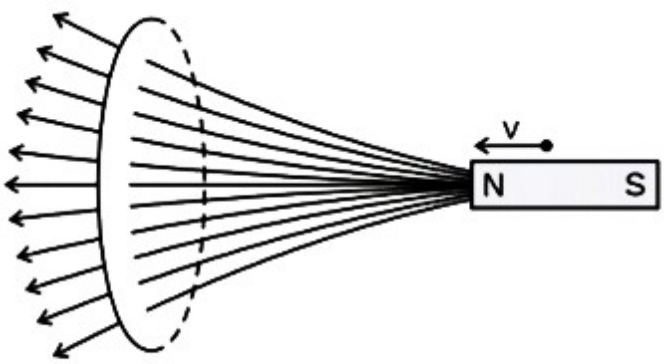
Lenz's Law

According to this direction of induce emf and induce current will be always such that it will oppose its' cause (that is, it will oppose the change of magnetic flux)
If ϕ increases $\Rightarrow$ B due to induced current will try to reduce magnetic flux
If ϕ decreases $\Rightarrow$ B due to induced current will try to increase the magnetic flux

Lenz's Law as Conservation of Energy Principle

Lenz's law suggests that induced emf opposes the cause which produces it.

In the given figure, the north pole of a magnet is approaching towards a metallic ring. For a bar magnet, the magnetic field lines come out from its north pole and the magnetic field intensity decreases with distance from the magnet.



So, as the magnet approaches the closed ring, the magnetic flux linked with the coil increases. To oppose this increase in magnetic flux, there will be an induced current in the coil which produces a magnetic field in opposite direction (towards right). For a magnetic field pointing towards right, the induced current in the ring should be anticlockwise as seen from the magnet's side.

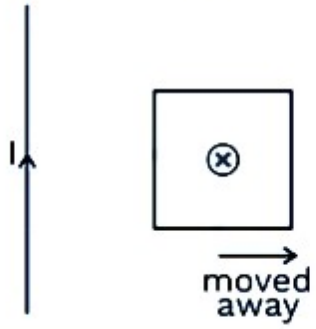
Another explanation can be given as the north pole of the magnet approaches the ring, its front face will act as a north pole to repel the magnet's north pole. Now, in order to act as a north pole, the current in the ring should be anticlockwise as seen from magnet's side.

If the cause of the change in flux is considered to be the approach of the magnet, then as per Lenz's law, a force in the direction opposite to the motion of the magnet acts on the magnet. If the ring is free to move, then it moves away from the magnet to reduce the magnetic flux.

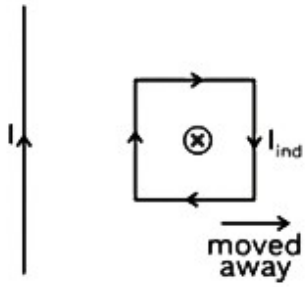
If the magnet is given some initial velocity towards the coil and is released, it will slow down. It can be explained as the following.

The current induced in the coil will produce heat. From the energy conservation, if heat is produced there must be an equal decrease of energy in some other form, here it is the kinetic energy of the moving magnet. Thus the magnet must slow down. So we can justify that the **Lenz's law is conservation of energy principle**.

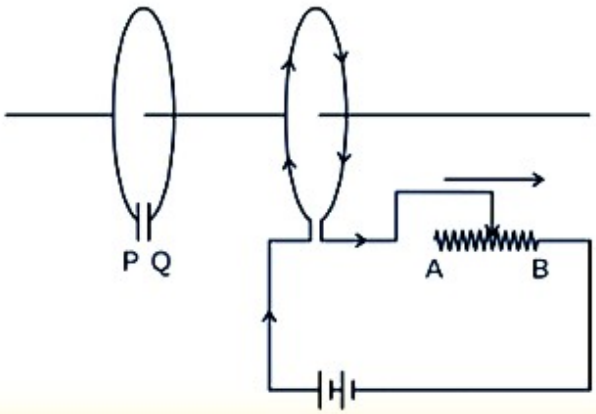
Q. **If the coil is moved away from infinitely long current carrying wire then find the direction of induced current in the loop?**



Sol. ⇒ Magnetic field due to wire is inward
⇒ ϕ linked with square is decreased
⇒ B due to induced current will try to increase flux
⇒ Current in the loop must be clockwise.

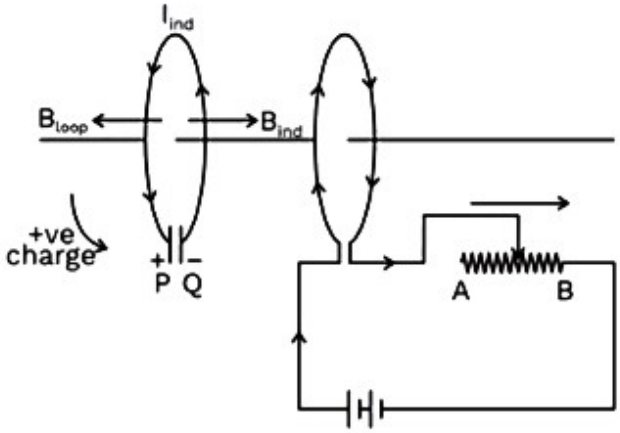


Q. **Find nature of charges appearing on plates P & Q, when resistance of rheostat is decreased**



Sol. When resistance of rheostat is decreased then current in loop will increase and hence flux associated with other loop will also increase. So current will be induced as shown in figure.

P = +ve
Q = –ve



Average Induced Current and Average Induced emf in a Coil

$$\varepsilon = -\frac{d\phi}{dt}$$

$$i_{ind} = \frac{\varepsilon}{R} = -\frac{1}{R}\frac{d\phi}{dt}$$

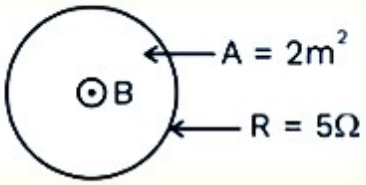
$$\int_{t_i}^{t_f} i_{in}\,dt = -\int_{\phi_i}^{\phi_f} \frac{1}{R}\,d\phi$$

Charge flown $= -\frac{1}{R}(\phi_f - \phi_i) \Rightarrow \boxed{\Delta q = -\frac{\Delta\phi}{R}}$

$$\frac{\Delta q}{\Delta t} = -\frac{\Delta\phi}{\Delta t}\frac{1}{R} \Rightarrow \boxed{I_{av} = -\frac{1}{R}\frac{\Delta\phi}{\Delta t}}$$

$$\boxed{\varepsilon_{av} = -\frac{\Delta\phi}{\Delta t}}$$

Q. **The magnetic field acting outside the plane of the coil in figure is decreasing at a rate of 10 T/s. What is the magnitude and direction of the induced current in the coil.**

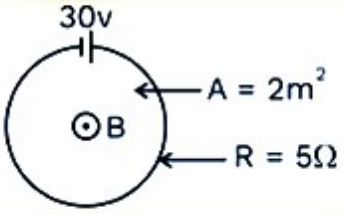


Sol. $\phi = B.A \Rightarrow \text{emf} = A \cdot \frac{dB}{dt} = 2 \times 10 = 20\ v$

$\therefore$ i = 20/5 = 4 amp.

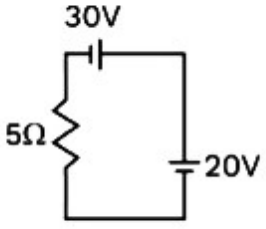
As per Lenz's law, direction of induced current will be anticlockwise.

Q. **The magnetic field acting outside the plane of coil is decreasing at a rate of 10 T/s. What is the magnitude and direction of the induced current if there is a 30 V battery is connected as shown?**

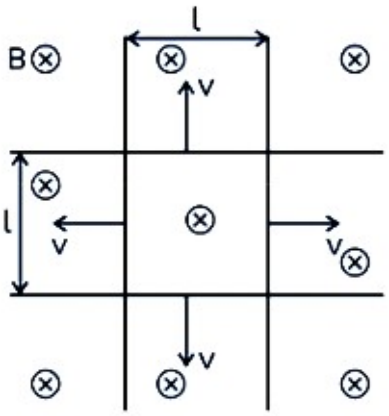


Sol. Induce emf = 20V

Equivalent I = 2A clockwise



Q. **Find induced emf in loop at the moment when side of loop is 'l'. If resistance per unit length is λ, also find induced current.**



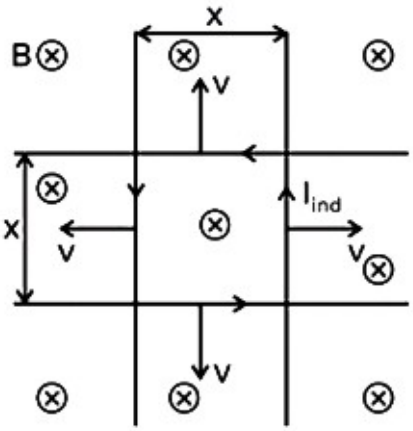
Sol. $\phi = Bx^2$

$$\varepsilon = -\frac{d\phi}{dt} = -2Bx\frac{dx}{dt}\left(\because \frac{dx}{dt} = 2v\right)$$

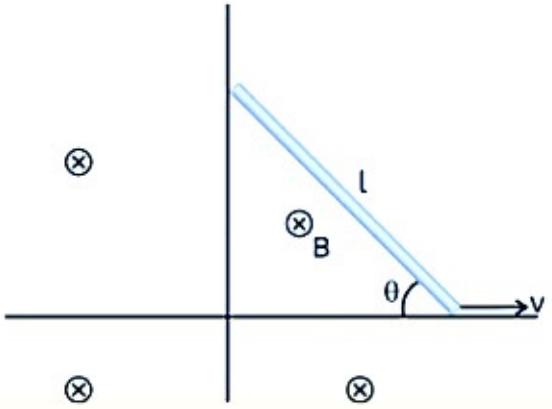
$\varepsilon = -4Bxv$

When $x = l \Rightarrow \varepsilon = -4Blv \Rightarrow |\varepsilon| = 4Blv$

$$I_{ind} = \frac{|\varepsilon|}{R} = \frac{4Bvl}{4\lambda l} = \frac{Bv}{\lambda}$$



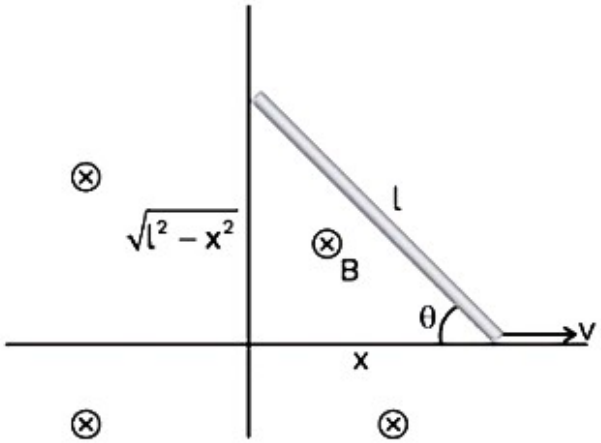
Q. **A rod is sliding between two rails as shown in the figure, find induced emf in the loop at given instant.**



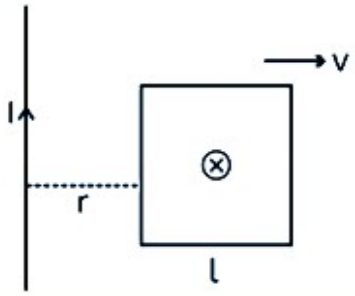
Sol. $\phi = B\frac{1}{2}x\sqrt{l^2 - x^2}$

$$\varepsilon = -\frac{d\phi}{dt} = -\frac{B}{2}\left[\sqrt{l^2 - x^2} + \frac{x(-2x)}{2\sqrt{l^2 - x^2}}\right] \times \frac{dx}{dt}$$

$$\Rightarrow \qquad \varepsilon = -\frac{B}{2}\left[l\sin\theta - \frac{(l\cos\theta)^2}{l\sin\theta}\right]v$$



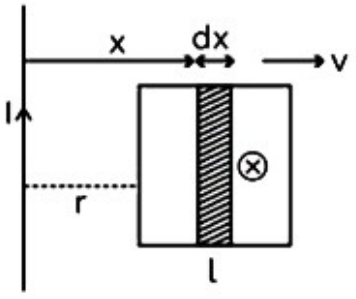
Q. **Find induced emf in the square loop as a function of I, l, v, r**



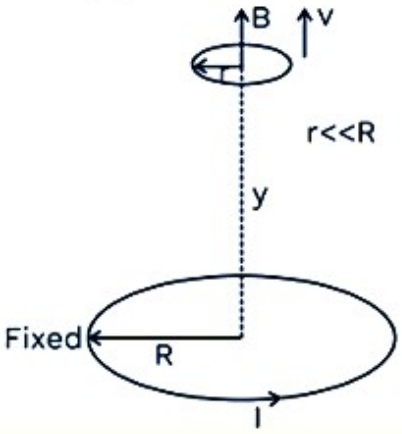
Sol. $\phi = \int_r^{r+l} \frac{\mu_0 I}{2\pi x} l\,dx = \frac{\mu_0 Il}{2\pi}\frac{\ln(r+l)}{r} \Rightarrow \qquad \phi = \frac{\mu_0 Il}{2\pi}\left(\ln\left(1+\frac{l}{r}\right)\right)$

$$\varepsilon = -\frac{d\phi}{dt} = -\frac{\mu_0 Il}{2\pi}\frac{1}{1+\frac{l}{r}}\left[0 - \frac{1}{r^2}\frac{dr}{dt}\right]$$

$$\varepsilon = +\frac{\mu_0 Il}{2\pi}\frac{lv}{(r+l)r} = \frac{\mu_0 Il^2 v}{2\pi r(l+r)}$$



Q. **Find induced emf in smaller ring (assume B to be uniform in ring)**



Sol.

$$\phi = BA = \frac{\mu_0 I R^2}{2(R^2 + y^2)^{3/2}} \times \pi r^2$$

$$\varepsilon = -\frac{d\phi}{dt} = -\frac{\mu_0 I R^2 r^2}{2}\left(\frac{-3}{2}\right)\left(\frac{1}{(R^2 + y^2)^{5/2}}\right)\left(2y\frac{dy}{dt}\right)$$

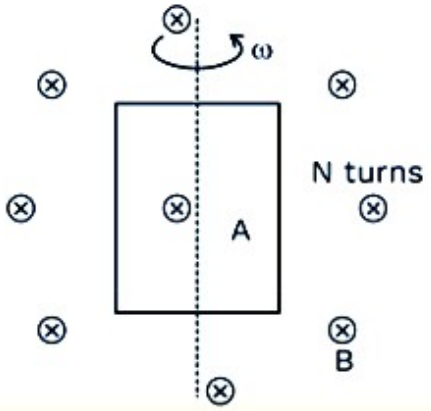
$$= \frac{3\mu_0 I R^2 r^2}{4(R^2 + y^2)^{5/2}} 2y\ v$$

Q. **At beginning plane of the coil is perpendicular to the magnetic field. If the coil start revolving with 'ω' angular velocity**
Find the following:

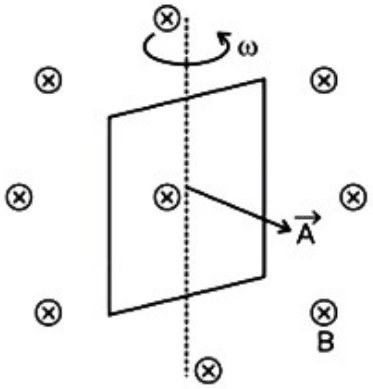
(i) Induced emf as a function of time

(ii) If resistance of coil is 'R', then heat produced in the coil in one complete revolution

(iii) Charge flown through the coil in half revolution



Sol. At t = t, coil rotates by an angle, $\theta = \omega t$



(i) $\phi = N(\vec{B} \cdot \vec{A})$

$= NBA\cos(\omega t)$

$\varepsilon = -\frac{d\phi}{dt} = -NBA(-\omega \sin \omega t)$

$= NBA\omega \sin\omega t$

(ii) $i = \frac{NBA\omega \sin \omega t}{R}$

Rate of heat produced $= i^2R = \frac{N^2B^2A^2\omega^2 \sin^2 \omega t}{R}$

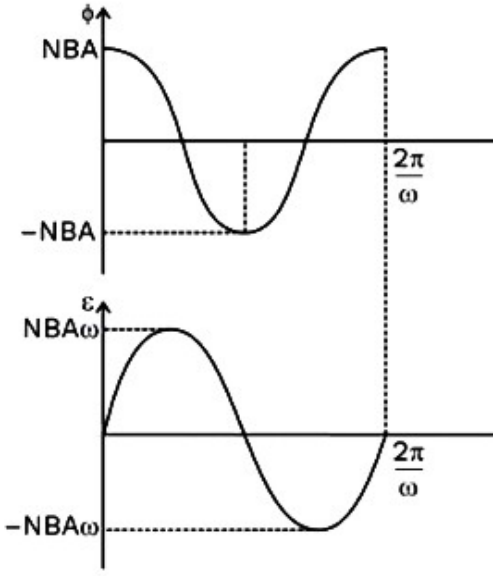
$\text{Heat} = \int i^2 R dt$

$\text{Heat} = \int_0^{\frac{2\pi}{\omega}} \frac{N^2B^2A^2\omega^2 \left(\sin^2 \omega t\right)(dt)}{R}$

$\cos 2\omega t = 1 - 2\sin^2\omega t \Rightarrow \sin^2 \omega t = \frac{1-\cos 2\omega t}{2}$

$\text{Heat} = \frac{N^2B^2A^2\omega^2}{R}\left(\frac{t}{2} - \frac{\sin 2\omega t}{2 \times 2\omega}\right)_0^{\frac{2\pi}{\omega}} = \frac{N^2B^2A^2\omega^2}{R} \times \frac{2\pi}{2\omega}$

$\text{Heat} = \frac{N^2B^2A^2\omega\pi}{R}$



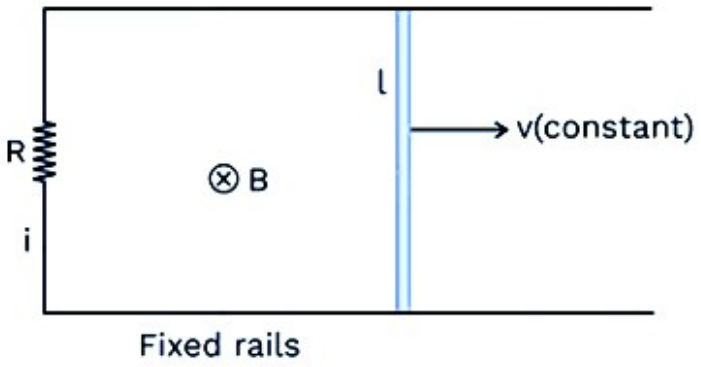
(iii) Charge flow Δq in half cycle $= -\frac{\Delta\phi}{R} = \frac{\phi_i - \phi_f}{R}$

$= \left(\frac{NBA + NBA}{R}\right)$

$= \frac{2NBA}{R}$

Charge flow in 1 cycle = 0

Q. **A conducting rod of length 'l' is pulled with constant speed 'v'. Neglect any resistance except 'R'.**



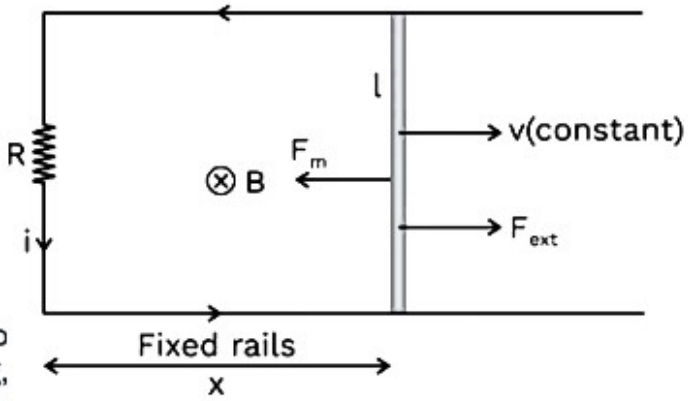
(i) Find induced current in the circuit
(ii) Force require to pull rod with constant velocity and its power
(iii) The rate of heat generated in the resistance.

Sol. (i) $\phi = Bxl$

$$\frac{d\phi}{dt} = Bl\frac{dx}{dt} = Blv$$

$$|\varepsilon| = \left|-\frac{d\phi}{dt}\right| = Blv$$

$$i = \frac{Blv}{R}$$



Because of v, flux due to magnetic field is increasing, hence induced current is in anticlockwise sense.

(ii) $F_m = IlB = \frac{B^2l^2v}{R}$

As v is constant $\Rightarrow F_{ext} = F_m = \frac{B^2l^2v}{R}$

$$P_{ext} = FV = \frac{B^2l^2v^2}{R}$$

(iii) $P = I^2R = \frac{B^2l^2v^2}{R}$

Rate of heat generated in resistor = power supplied by external force.

Motional emf

Motional emf

When a conductor is moved in magnetic field, free electrons of conductor will experience magnetic force because of this some charge is induced on surface of conductor. These induced charges will produce electric field in conductor. Deposition of charge will continue until the net force on free electrons becomes zero. Because of the induced charges, some potential difference is developed between two points of conductor, this potential difference is called motional emf

$\vec{F}_E + \vec{F}_m = 0$

$(-e)\vec{E} + (-e)(\vec{V} \times \vec{B}) = 0$

$\vec{E} = -(\vec{V} \times \vec{B}) \Rightarrow$ direction of electric field is opposite to $\vec{V} \times \vec{B}$

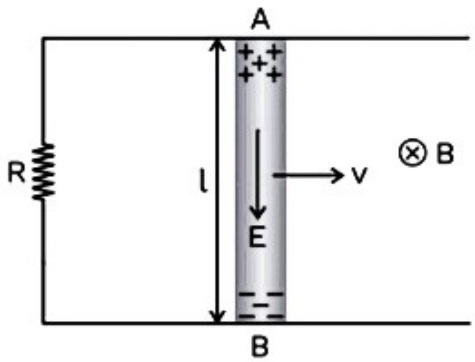
$$V_B - V_A = -\int_A^B \vec{E} \cdot \overrightarrow{dl}$$

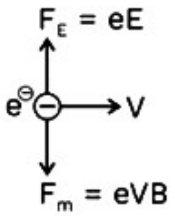
If $\vec{E}$ = uniform

$V_B - V_A = -\vec{E} \cdot \vec{d} \quad (\vec{d} = \vec{r}_B - \vec{r}_A)$

$$\boxed{V_B - V_A = (\vec{V} \times \vec{B}) \cdot \vec{d}}$$

Potential increases if we move in the direction of $\vec{V} \times \vec{B}$

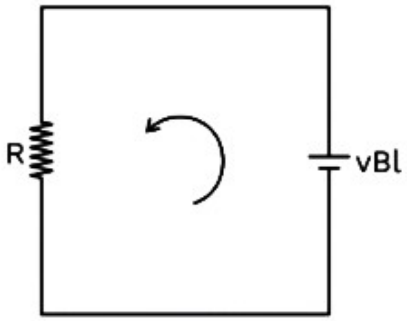




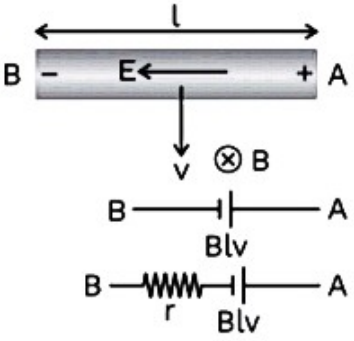
eE = eVB

E = VB

$V_A - V_B$ = EL = VBl



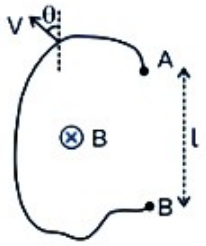
If a conducting rod is moving in a magnetic field such that its velocity, its length and magnetic field all three are perpendicular to each other, then



Electrical equivalent of rod if it has no resistance

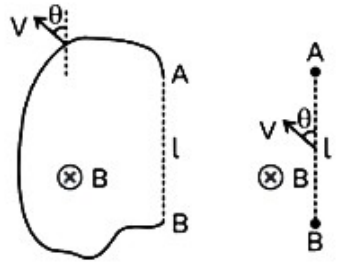
Electrical equivalent of rod if it has a resistance 'r'

Q. **Figure shows an irregular shaped wire AB moving with velocity v, as shown**

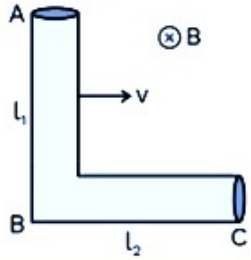


Find the emf induced in the wire.

Sol. Join the end points of the irregular wire and make it a straight one. Emf induced will be same in both cases which is Bvl sinθ



Q. **An L shaped conducting road is moving with velocity V in a magnetic field of strength B, as shown in figure. Find $V_A - V_B$ and $V_B - V_C$.**



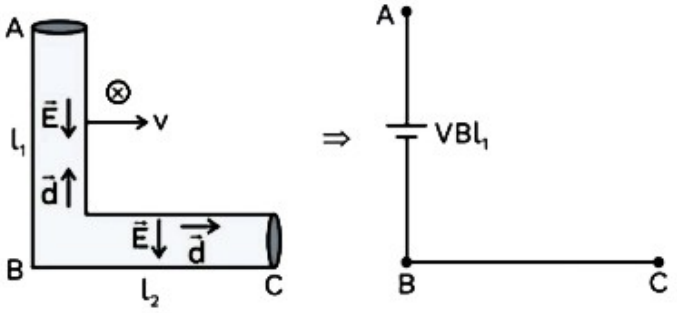
Sol. Direction of electric field is opposite to $\vec{V} \times \vec{B}$

$\vec{E} = -(\vec{V} \times \vec{B}) \Rightarrow E = |\vec{V} \times \vec{B}| = VB\sin 90° = VB$

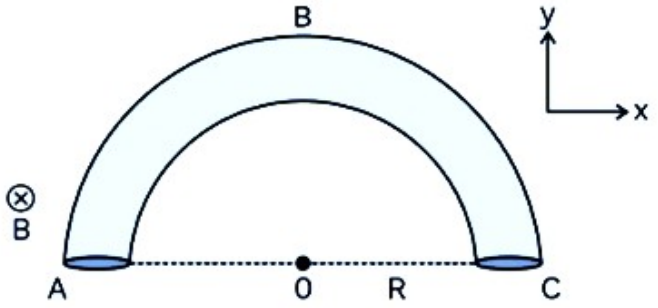
$V_A - V_B = -\vec{E} \cdot \vec{d} = -El_1 \cos 180°$

$V_A - V_B = -(VB)l_1(-1) = VBl_1$

$V_B - V_C = -\vec{E} \cdot \vec{d} = -El_2 \cos 90° = 0$



Q. **A conducting semi-circular rod of radius R is moving with velocity $\vec{V} = u\hat{i} + u\hat{j}$**



Find $V_A - V_B$, $V_B - V_C$ and $V_A - V_C$.

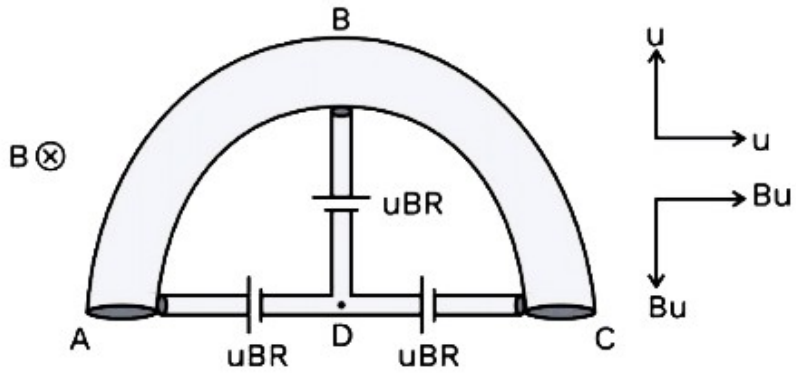
Sol. $\vec{V} = u\hat{i} + u\hat{j}$, $\vec{E} = -(\vec{V} \times \vec{B}) = \vec{B} \times \vec{V}$

$\vec{B} \times \vec{V} = (-B\hat{k}) \times (u\hat{i} + u\hat{j}) = Bu\hat{i} - Bu\hat{j}$

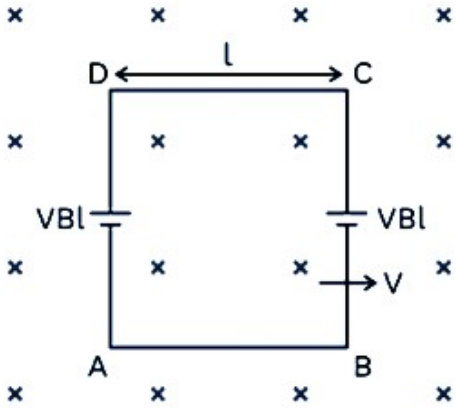
$V_A - V_B = 0$

$V_B - V_C = 2\ uBR$

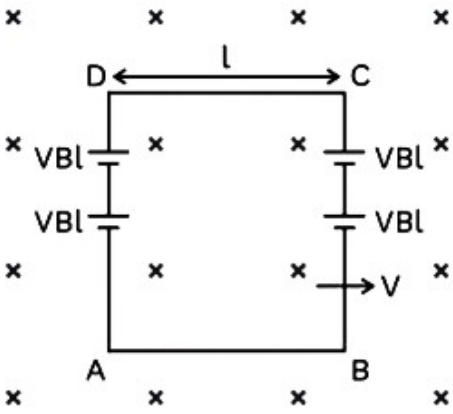
$V_A - V_C = 2\ uBR$

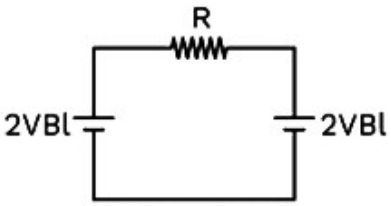


Q. **A square loop of side length L and resistance R contains two voltage sources of emf VBl and moves with velocity V in a magnetic field of strength B as shown in figure. Find current in the loop.**



Sol.



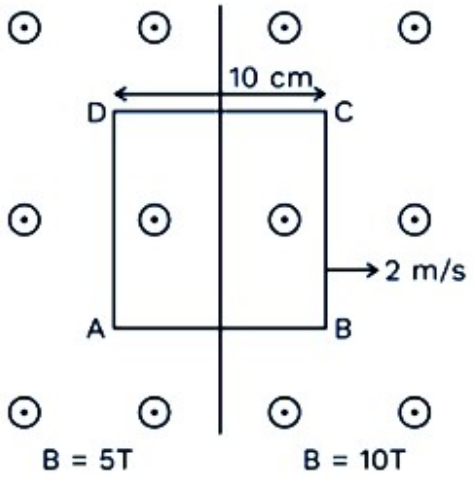


No current in loop as both batteries cancel out the total emf in loop.

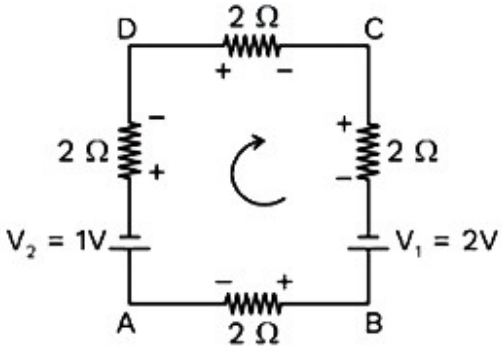
Q. **ABCD is a square loop of side length 10 cm.**
Resistance per unit length = 20 Ω/m.

(1) Find the potential difference between points (i) A & B, (ii) B & C

(2) Force required to pull it with constant velocity



Sol.



(1) $V_1 = 5 \times 2 \times 0.1 = 1V$ and $V_2 = 10 \times 2 \times 0.1 = 2V$

$$i = \frac{2-1}{8} = \frac{1}{8}A$$

(i) $$\left|V_A - V_B\right| = 2 \times \frac{1}{8} = \frac{1}{4}V \Rightarrow V_B - V_A = \frac{1}{4}V$$

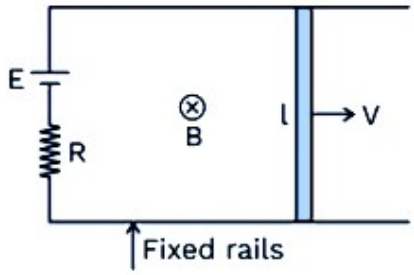
(ii) $$V_B - V_C = 2 - \frac{1}{8} \times 2 = \frac{4 \times 2}{4} - \frac{1}{4} = \frac{7}{4}V$$

(2) $$F_{BC} = I(\vec{l} \times \vec{B}) = \frac{1}{8}(0.1)(10) \text{ (towards left),}$$

$$F_{AD} = \frac{1}{8}(0.1)(5) \text{ (towards right)}$$

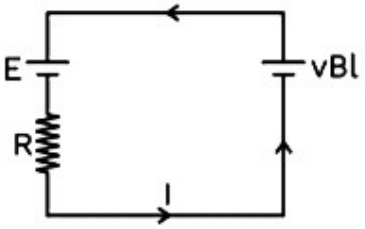
$$F_{BC} - F_{AD} = \frac{1}{8}(0.1)(5) = \frac{0.5}{8} = \frac{5}{80} = \frac{1}{16}N \text{ (towards left)}$$

Q. **What should be current in R?**



Sol. $I = \frac{vBl - E}{R}$

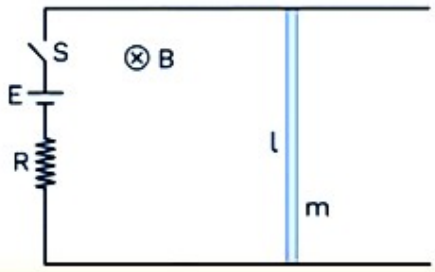
If $v = \frac{E}{Bl}$, then E & vBl will cancel each other, no current will flow in the loop. Hence no force on conducting rod 'l' and it will continue to move with $v = \frac{E}{Bl}$



If $v > \frac{E}{Bl}$, vBl dominates $\Rightarrow$ current in circuit is in anticlockwise direction $\Rightarrow$ force on conducting rod is leftward $\Rightarrow$ v decreases and becomes $v = \frac{E}{Bl}$ (constant) after some time.

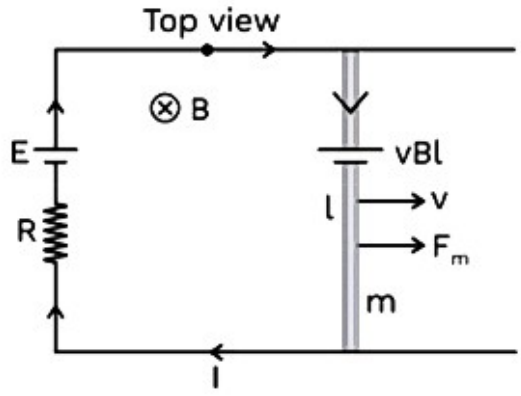
If $v < \frac{E}{Bl}$, E dominates $\Rightarrow$ current in circuit is in clockwise direction $\Rightarrow$ force on conducting rod is rightward $\Rightarrow$ v increases and becomes $v = \frac{E}{Bl}$ (constant) after some time.

Q. **A conducting rod of length l and mass m if placed on horizontal fixed rails whose top view is shown in figure. Now switch S is closed. Find velocity of rod as a function of time.**



Sol. Diagram at any time t is shown.

$F = IlB$



$$F = \left(\frac{E - vBl}{R}\right)lB$$

$$m\frac{dv}{dt} = \left(\frac{E - vBl}{R}\right)lB \Rightarrow m\int_0^v \frac{dv}{E - vBl} = \int_0^t \frac{lBdt}{R}$$

$$\left[\frac{\ln(E - vBl)}{-Bl}\right]_0^v = \frac{lB(t)}{mR}$$

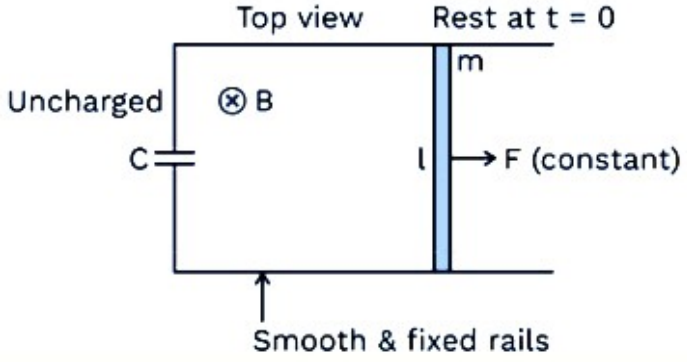
$$\ln\left(\frac{E - vBl}{E}\right) = -\frac{l^2B^2t}{mR}$$

$$\frac{E - vBl}{E} = e^{-\frac{l^2B^2t}{mR}} \Rightarrow E - Ee^{-\frac{l^2B^2l}{mR}} = vBl$$

$$V = \frac{E}{Bl} - \frac{E}{Bl}e^{-\frac{l^2B^2l}{mR}}$$

$$v = \frac{E}{Bl}\left(1 - e^{-\frac{l^2B^2t}{mR}}\right)$$

Q. **A conducting rod of length l and mass m if placed on horizontal fixed rails whose top view is shown in figure. A constant force F is applied on conducting rod. Find velocity as a function of time.**



Sol. Let at any instant velocity of rod is v and charge q has flown through the circuit.

$\frac{q}{c} = vBl$ (ΔV across rod & capacitance should be same as they are in parallel)

$Q = vBlC$

$I = \frac{dq}{dt} = BlC\frac{dv}{dt} = BlCa$

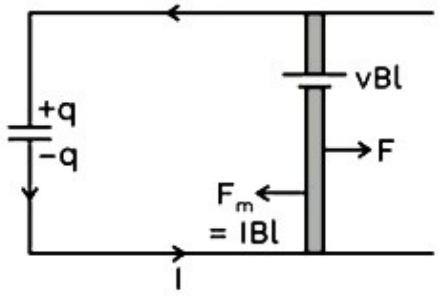
$F_m = IlB = B^2l^2Ca$

$F - B^2l^2Ca = ma$

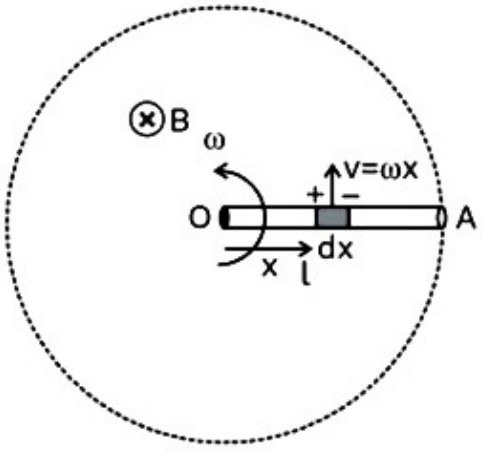
$\Downarrow$

$a = \frac{F}{m + B^2l^2C}$

$v = \left(\frac{F}{m + B^2l^2C}\right)t$



Emf Induced in a Rotating rod Placed in Uniform Magnetic Field



A conducting rod of length l is rotating with constant angular speed ω such that magnetic field is perpendicular to its plane of rotation.

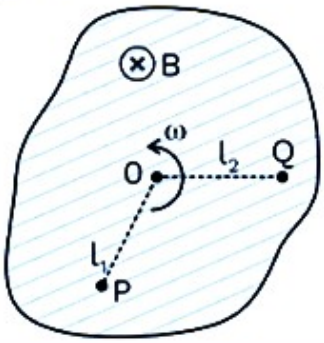
$dv = vBdx$

$v = \int_0^l \omega x B dx$

$v = \omega B\left(\frac{l^2}{2}\right)$

$$\boxed{v_0 - v_A = \frac{B\omega l^2}{2}}$$

Q. **A conductor as shown in figure is rotating with constant angular speed ω such that magnetic field is perpendicular to its plane of rotation. Find potential difference between points Q and P.**

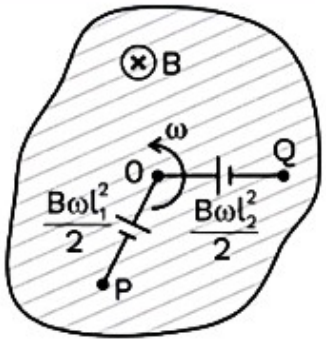


Sol.

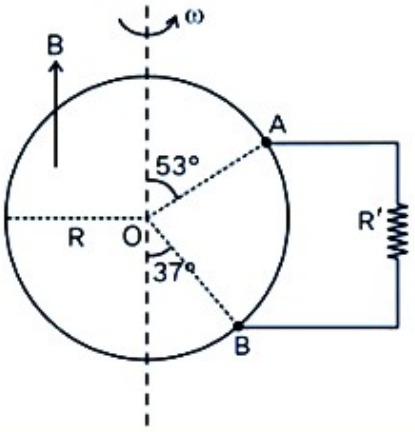
$$v_O - v_P = \frac{B\omega l_1^2}{2}$$

$$v_O - v_Q = \frac{B\omega l_2^2}{2}$$

$$v_Q - v_P = \frac{B\omega}{2}(l_1^2 - l_2^2)$$



Q. **A conducting sphere of radius R is rotating in a uniform magnetic field of strength B such that its axis of rotation is parallel to direction of magnetic field. Find current through a resistance R′ connected between points A and B as shown in figure.**



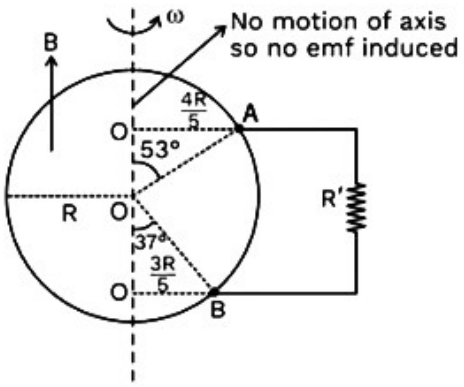
Sol.

$$v_A - v_0 = \frac{B\omega}{2}\left(\frac{4}{5}R\right)^2$$

$$v_B - v_0 = \frac{B\omega}{2}\left(\frac{3}{5}R\right)^2$$

$$v_A - v_B = \frac{B\omega R^2}{2(25)}(7)$$

$$i = \frac{7B\omega R^2}{50R'}$$

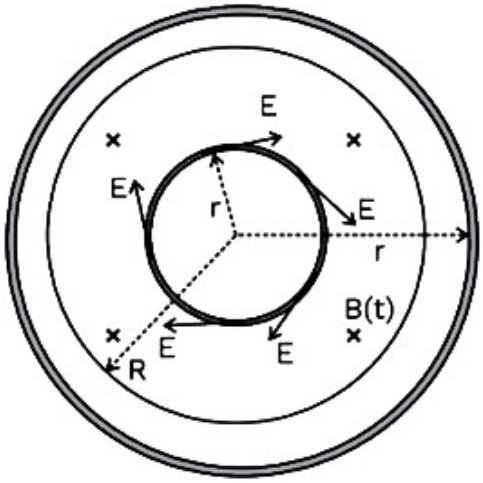


Induced Electric Field

Induced Electric Field

Consider a uniform magnetic field in cylindrical region of radius 'R' as shown in the figure. Magnetic field changes with time. If we place a circular ring of radius small 'r', coaxially then some emf will be induced in this coil. Origin of this induced emf is due to induced electric field. Induced electric field is produced by time varying (changing) magnetic field.

If $r \leq R$



Induced emf $\oint \vec{E}.\overrightarrow{dl} = -\frac{d\phi}{dt}$

$$\int E dl \cos 0^\circ = -\frac{d}{dt}(B\pi r^2)$$

$$E2\pi r = -\pi r^2 \frac{dB}{dt}$$

$$\boxed{E = -\frac{r}{2}\frac{dB}{dt}}$$

If $r \geq R$

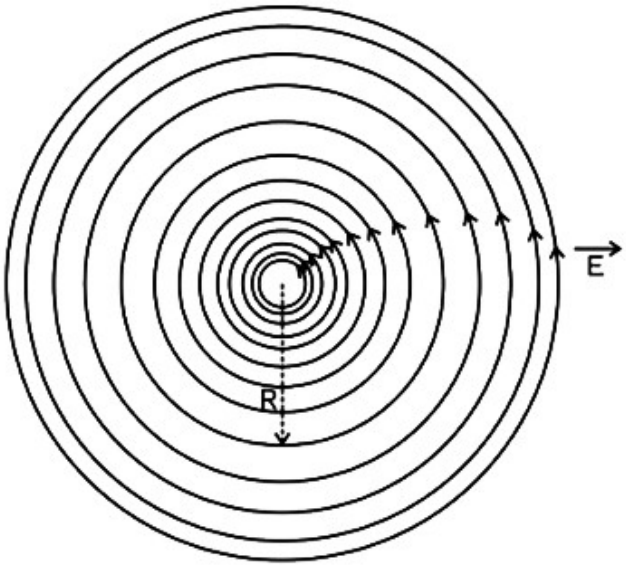
Then $\oint \vec{E}.\overrightarrow{dl} = -\frac{d\phi}{dt}$

$$\int E dl \cos 0^\circ = -\frac{d}{dt}(B\pi R^2)$$

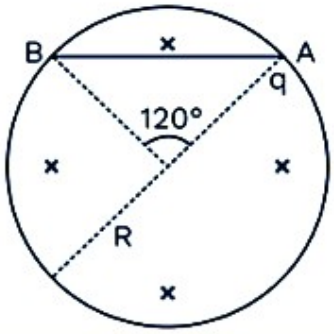
$$E2\pi r = -\pi R^2 \frac{dB}{dt}$$

$$\boxed{E = -\frac{R^2}{2r}\frac{dB}{dt}}$$

Electric field lines due to induced electric field form close loop. Work done by induced electric field for a cyclic path is not necessarily to be zero. Work done by induced electric filed in moving charge from one point to another point depends on path taken, that is why it is non-conservative field.



Q. **A point charge 'q' is moved from. A to B along straight chord. If magnetic field in region is given by B = kt, find work done by induced electric field in this process.**

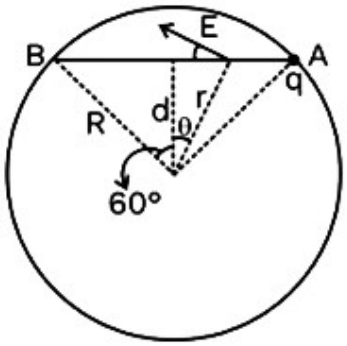


Sol.

$$E = -\frac{r}{2}\frac{dB}{dt} = -\frac{rk}{2}$$

$$W = \int_A^B -qE\cos\theta dl = \int_A^B q\frac{rk}{2}\cos\theta dl$$

$$= \int_A^B q\frac{rk}{2}\frac{d}{r}dl = q\frac{kd}{2}R\sqrt{3}$$



Or

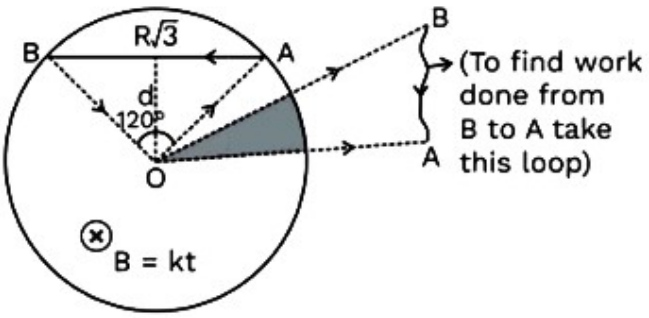
Induced emf = work done by induced electric field in moving a unit positive test charge round the loop

$$\int_{ACBA} \vec{E}\cdot\overrightarrow{dl} = -\frac{d\phi}{dt}$$

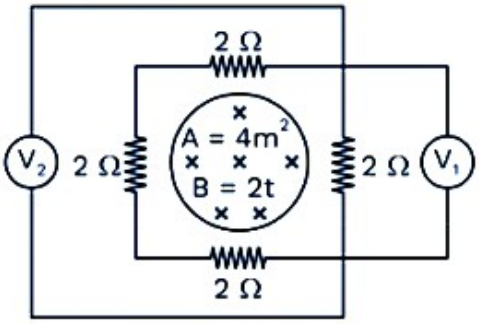
$$q\int_A^O \vec{E}\cdot\overrightarrow{dl} + q\int_O^B \vec{E}\cdot\overrightarrow{dl} + q\int_B^A \vec{E}\cdot\overrightarrow{dl} = -\frac{d\phi}{dt}q$$

$$0 + 0 + q\int_B^A \vec{E}\cdot\overrightarrow{dl} = -q\frac{R\sqrt{3}\,dk}{2}$$

$$q\int_A^B \vec{E}\cdot\overrightarrow{dl} = q\frac{R\sqrt{3}\,dk}{2}$$



Q. **Find reading of voltmeters V_1 & V_2**



Sol. $\phi = 8t$

$$\varepsilon = -\frac{d\phi}{dt} = -8$$

$$\frac{\varepsilon}{R} = \frac{8}{8} = 1A \text{ (current in the loop CGHD)}$$

Writing kvl along path ABCD

$\Rightarrow \quad -V_1 + 2 = 0$

$V_1 = 2V$

Writing kvl along path EFCGHD

$\Rightarrow \quad +V_2 - 2 - 2 - 2 = 0$

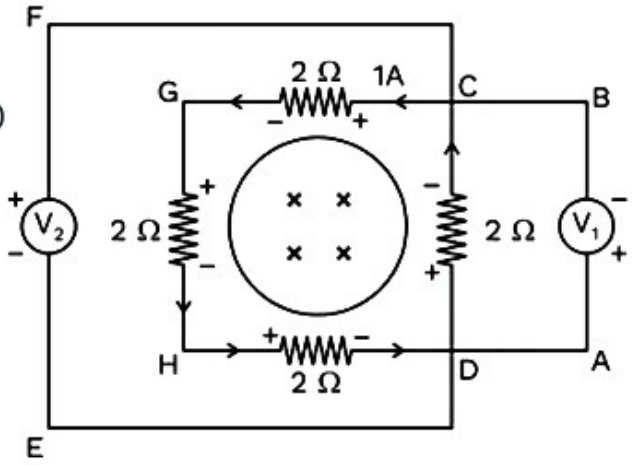
$V_2 = 6v$

Writing kvl along path ABCGHD

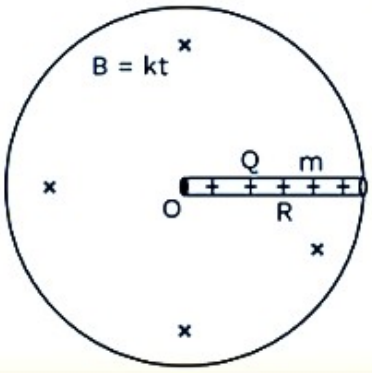
$\Rightarrow \quad -V_1 - 2 - 2 - 2 + (8) = 0$

(8) → Induced emf

$V_1 = 2v$



Q. **A non-conducting rod of mass m having charge Q uniformly distributed over its length R is placed in a time varying magnetic field (B = kt) and rigidly fixed at point O as shown in figure. Find angular acceleration of rod.**

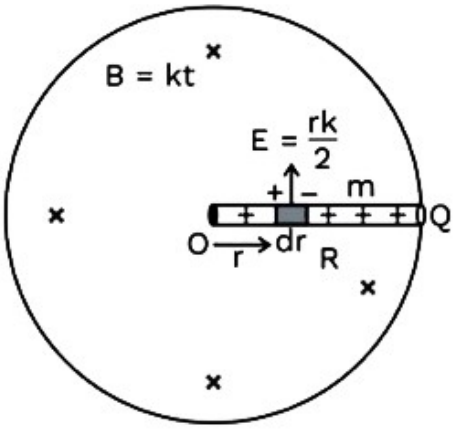


Sol. $E = \frac{r}{2}\frac{dB}{dt} = \frac{r}{2}k$

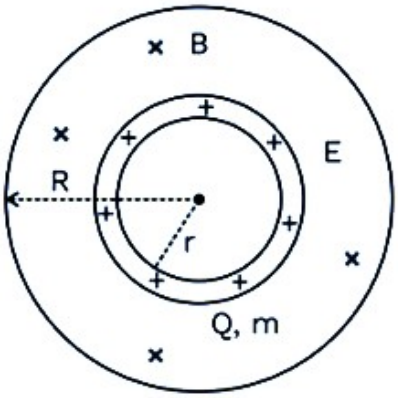
$dq = \frac{Q}{R}dr$

$I\alpha = \int_0^R \left(\frac{Q}{R}dr \cdot \frac{r}{2}k\right)r$

$\frac{mR^2}{3} \cdot \alpha = \frac{Qk}{2R}\frac{R^3}{3} \Rightarrow \alpha = \frac{Qk}{2m}$



Q. **A non-conducting ring of mass m and uniformly distributed charged q is placed in a magnetic field B as shown in figure. If we turn off magnetic field, then find 'ω' acquired by ring.**



Sol. $I\alpha = \tau = EQr$

$\frac{Id\omega}{dt} = \left(\frac{-r}{2}\frac{dB}{dt}\right)Qr$

$\int_0^\omega Id\omega = \int_B^0 \left(\frac{-r}{2}dB\right)Qr$

$I(\omega - 0) = \frac{-r^2Q}{2}[B]_B^0$

$\omega = \frac{-r^2Q}{2I}(-B) \Rightarrow \omega = \frac{Qr^2B}{2I} = \frac{Qr^2B}{2mr^2} = \frac{QB}{2m}$

Eddy Current and Electromagnetic Damping

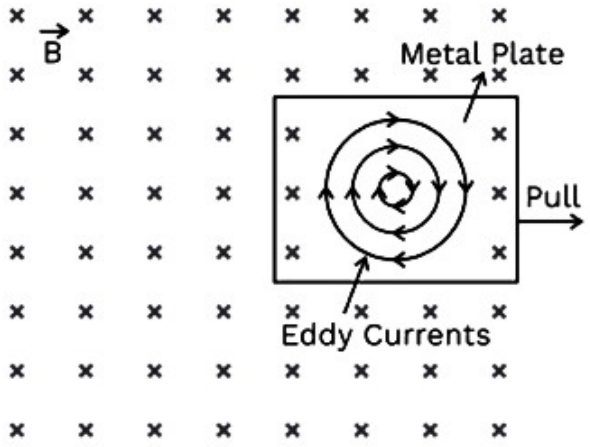
Eddy Current

When a conductor is placed in a changing magnetic field, induced emf is produced in it. As a result local currents are produced in the conductor. These local currents are called eddy currents.

If a conducting material is moved in a magnetic field, then eddy currents are also produced. Eddy currents flow in closed paths.

There is loss of energy due to eddy currents and it appears in the form of heat.

In order to minimize the energy loss in the form of heat due to eddy currents the core of dynamo, motor or transformer is not taken as a single piece of soft iron but in the form of a pack of thin sheets insulated from each other by a layer of insulating varnish, called laminated core. This device increases the resistance for the eddy currents. In this way eddy currents are considerably reduced and loss of energy becomes less.



Uses of eddy currents :

(a) Moving coil galvanometer
(b) Induction furnace
(c) Dead beat galvanometer
(d) Speedometer
(e) Electric brakes

Electromagnetic Damping

In controlling the motion of an object Damping plays an important role. Damping is opposition offered to the motion of an object. It always tends to reduce the speed of a moving object. Objects can possess various types of motion like rotatory motion, linear motion, oscillatory motion etc.

To name a few damping techniques which are used in rotating, moving and oscillating systems, which includes, fluid friction damping, air friction damping, conventional friction damping, and electromagnetic damping.

Among all techniques of damping, one of the most fascinating damping techniques is electromagnetic damping. Electromagnetic damping uses electromagnetically induced current to control/regulate/slow down the motion of the object without any actual physical contact with the moving object.

When a conductor moves through a magnetic field, the movement induces an eddy current in the conductor. When these Eddy currents and magnetic field interact with each other, a damping force is generated. The motion of the conductor/object is opposed by it. A damping technique where electromagnetically induced current slows down the motion of an object without any actual contact is known as the electromagnetic damping.

The electromagnetic damping force is proportional to the induced eddy current, strength of the magnetic field and the speed of the object. Which implies that greater will be the damping if faster the object moves, and lower will be damping if slower is the motion of object, which will result in the smooth stopping of the object.

Self Inductance

Self Inductance

When a current flowing through a coil changes with time, some emf is induced in the coil which opposes the change of current. This phenomena is called self inductance.

Consider a coil through which I current is flowing. Flux linked with the coil is proportional to current flowing through the coil.

$\phi \alpha I$

$$\boxed{\phi = LI}$$

L = self inductance

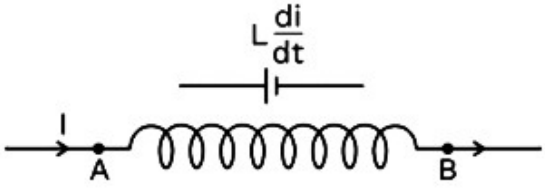
SI unit : $\frac{Wb}{A} = \frac{v \cdot s}{A} = \Omega \cdot s$ = Henry (H)

$$\varepsilon = \frac{-d\phi}{dt} = \frac{-d(LI)}{dt}$$

$$\boxed{\varepsilon = -L\frac{dI}{dt}}$$

$\Rightarrow$ Ideal inductor (resistance = 0)

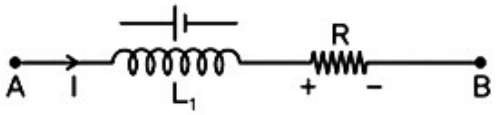
$$V_A - V_B = \frac{LdI}{dt}$$



$\Rightarrow$ Non ideal inductor (resistance $\neq$ 0)

$$V_A - \frac{LdI}{dt} - IR = V_B$$

$$V_A - V_B = \frac{LdI}{dt} + IR$$

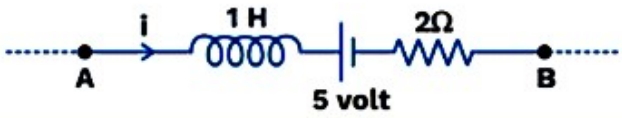


Q. **AB is a part of circuit. Find the potential difference $V_A - V_B$ if**

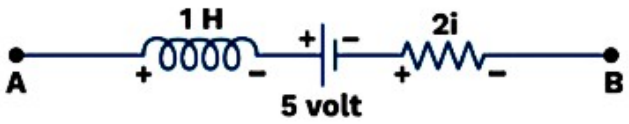
(i) current i = 2A and is constant

(ii) current i = 2A and is increasing at the rate of 1 amp/sec

(iii) current i = 2A and is decreasing at the rate 1 amp/sec



Sol. $L\frac{di}{dt} = 1\frac{di}{dt}$



Writing KVL from A to B

$$V_A - 1\frac{di}{dt} - 5 - 2i = V_B$$

(i) Put $i = 2, \frac{di}{dt} = 0$ $\Rightarrow$ $V_A - 5 - 4 = V_B$ $\therefore$ $V_A - V_B = 9$ volt

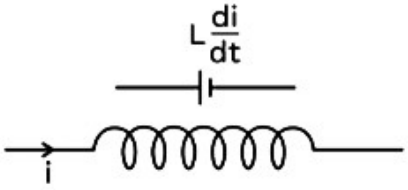
(ii) Put $i = 2, \frac{di}{dt} = 1$ $\Rightarrow$ $V_A - 1 - 5 - 4 = V_B$ or $V_A - V_B = 10$ volt

(iii) Put $i = 2, \frac{di}{dt} = -1$ $\Rightarrow$ $V_A + 1 - 5 - 2 \times 2 = V_B$ or $V_A = 8$ volt

Energy of Inductor

When we try to increase current in a coil or inductor, some emf is induced in the coil opposite to the current so some work is required to establish some current in the coil. This work is stored in the form of magnetic energy in field.

Consider a coil of self inductance 'L'. Initially current in the coil is zero. Suppose at any instant when we are increasing current, current in the coil is 'i'



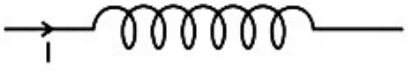
$$P = VI = L\frac{di}{dt}i$$

Work done in dt time (dW) = Pdt = Lidi

Total work done to established I current,

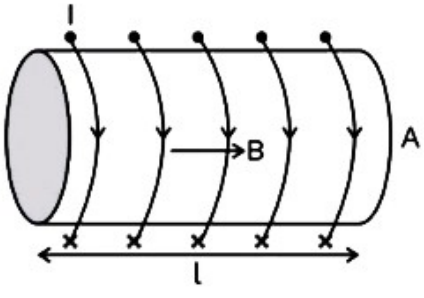
$$W = \int_0^I Lidi = \frac{1}{2}LI^2 = U$$

$$\boxed{U = \frac{1}{2}LI^2}$$



Self Inductance of an Ideal Solenoid:

Consider a solenoid with total no of turns 'N', length 'l', area of cross section 'A'. Assume that Its length is large in comparison to radius.



$$B = \mu_0 nI = \mu_0 \frac{N}{l} I$$

Magnetic flux linked with one turn $= BA = \frac{\mu_0 NIA}{l}$

Total flux $\phi = NBA = \frac{\mu_0 N^2 Al}{l}$

$\phi = LI$

$$\boxed{L = \frac{\mu_0 N^2 A}{l}}$$

L depends on
(i) Geometry, shape, size of solenoid
(ii) Medium (as μ of different mediums is different)
(iii) Number of turns

Magnetic Energy Density

Volume of solenoid, V = A × l

Magnetic energy stored in solenoid, $U = \frac{1}{2}LI^2 = \frac{1}{2}\left(\frac{\mu_0 N^2 A}{l}\right)I^2$

Magnetic energy density, $u = \frac{U}{V} = \frac{\frac{1}{2}\left(\frac{\mu_0 N^2 A}{l}\right)I^2}{Al} = \frac{1}{2\mu_0}\left(\mu_0 \frac{N}{l} I\right)^2 = \frac{B^2}{2\mu_0}$

Magnetic energy density, $\boxed{u = \frac{B^2}{2\mu_0}}$

LR Circuit

LR Ciruit

Q. **Consider an inductor of self inductance 5H (Henry). Initially current in the coil is zero. It is connected to a battery of emf 10 V and negligible resistance. Inductor is also ideal. Find current in inductor after 4 sec.**

Sol.

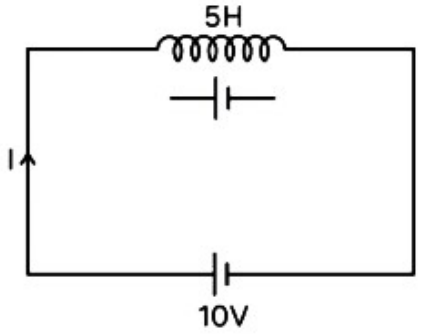
$$+10 - \frac{L dI}{dt} = 0$$

$$\frac{dI}{dt} = 2$$

$$\int_0^I dI = \int_0^4 2dt$$

$$I - 0 = 2(4)$$

$$I = 8A$$



Analogous Problems

$L \leftrightarrow m$, $\phi \leftrightarrow p$, $I \leftrightarrow v$, Emf $\leftrightarrow F$

$\frac{1}{2}Li^2 \leftrightarrow \frac{1}{2}mv^2$, $P = mv \leftrightarrow \phi = LI$

Self inductance is also called electrical inertia

Q. **Consider a block of mass 5 kg on a smooth floor pulled by a constant horizontal force of 10 N. Find the velocity of block at 4 sec**

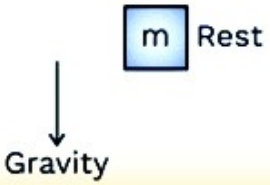
Sol.

$$10 = 5\frac{dv}{dt} \Rightarrow dv = 2dt \Rightarrow \int_0^v dv = \int_0^4 2dt$$

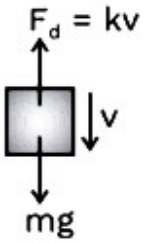
$$v = [2t]_0^4 = 8 \text{ m/s}$$



Q. **A block of mass m is released from rest under gravity. It experiences a drag force given by F = –kv. Find velocity of block as a function of time.**



Sol. $mg - kv = ma \Rightarrow mg - kv = m\frac{dv}{dt}$

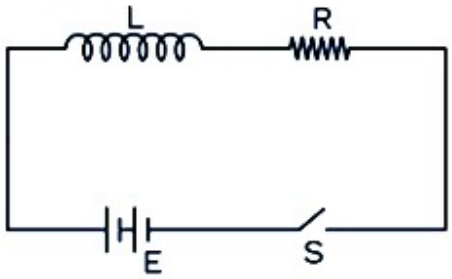


$$\int_0^t \frac{dt}{m} = \int_0^v \frac{dv}{(mg - kv)} \Rightarrow \frac{t}{m} = -\frac{1}{k}\ln\left(\frac{mg - kv}{mg}\right) \Rightarrow -\frac{kt}{m} = \ln\left(\frac{mg - kv}{mg}\right)$$

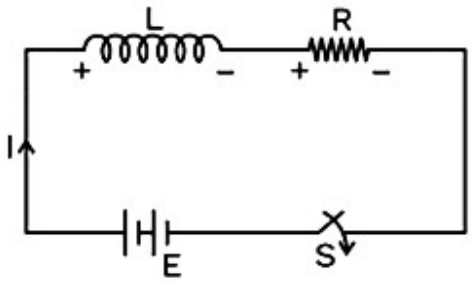
$$mge^{-\frac{kt}{m}} = mg - kv \Rightarrow v = \left(\frac{mg - mge^{-\frac{kt}{m}}}{k}\right)$$

$$v = \frac{mg}{k}\left(1 - e^{-\frac{kt}{m}}\right)$$

Q. **An inductor of inductance L is connected with resistor of resistance R and a battery of emf E as shown in figure. Find current through inductor as a function of time after closing switch 'S'.**



Sol.



At t = 0, I = 0

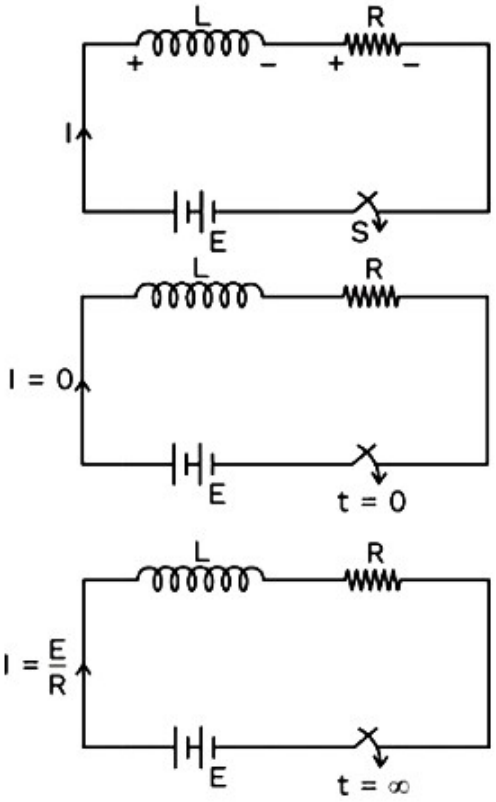
Writing KVL at $t = t \Rightarrow E - L\frac{dI}{dt} - IR = 0$

analogous to

$F - \frac{mdv}{dt} - vk = 0$ $(R \leftrightarrow k)$

$$I = \frac{E}{R}\left(1 - e^{-\frac{Rt}{L}}\right)$$

Current in LR Series Circuit Just After Closing Switch (t = 0) and Long Time After Closing Switch (Steady State or t → ∞)



At t = t

$$E - L\frac{dI}{dt} - IR = 0$$

At t = 0

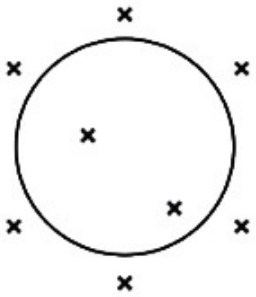
$$I = 0;\ \frac{dI}{dt} = \frac{E}{L}$$

$$V_{Resistance} = 0;\ V_{Inductor} = E$$

At t = ∞

$$\frac{dI}{dt} = 0,\ I = \frac{E}{R}$$

In steady state, the inductor can be treated as straight wire because potential difference across it is zero.

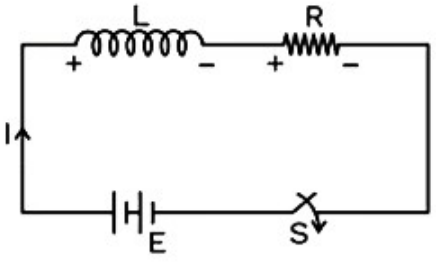


Magnetic flux linked with inductor cannot change instantaneously; it will change gradually if some voltage is applied cross it

If there is a coil of zero resistance (ideal inductor) and no emf is applied across it, then net magnetic flux through coil will remain constant.

$$\boxed{\phi = \vec{B}\cdot\vec{A} + LI = \text{constant}}$$ ⇒ Law of conservations of flux

Growth of Current in Series LR Circuit



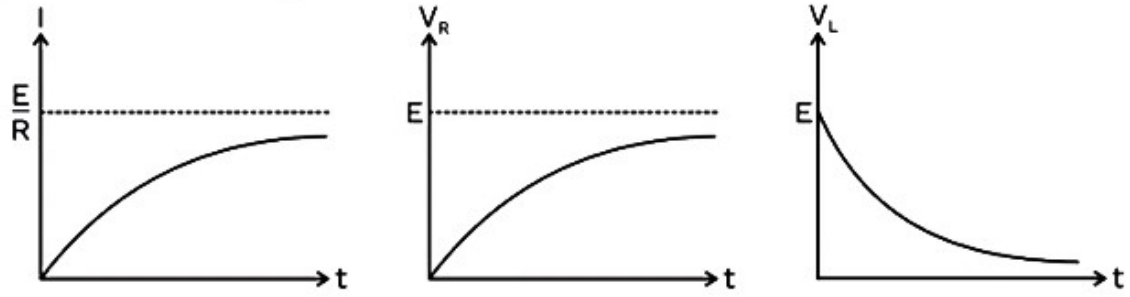
$$E - L\frac{dI}{dt} - IR = 0 \Rightarrow \int_0^I \frac{dI}{E - Ir} = \int_0^t \frac{dI}{L}$$

$$\left[\frac{\ln(E - IR)}{-R}\right]_0^I = \frac{t}{L} \Rightarrow E - IR = Ee^{-\frac{Rt}{L}}$$

$I = \frac{E}{R}\left(1 - e^{-\frac{Rt}{L}}\right)$, $\left(\text{Time constant } \tau = \frac{L}{R}\right)$

Voltage across resistor, $V_R = IR = E\left(1 - e^{-\frac{Rt}{L}}\right)$

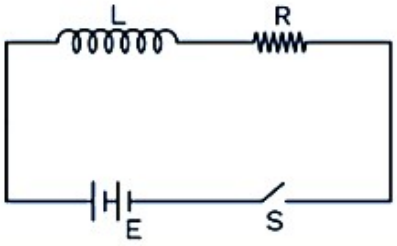
Voltage across inductor, $V_L = E - IR = Ee^{-\frac{Rt}{L}}$



Final current in circuit is $\frac{E}{R}$, which is independent of L. After one time constant $\left(\tau = \frac{L}{R}\right)$, current in circuit = 63% of the final current. More time constant of the circuit implies slower rate of change of current.

If there is any change in the circuit containing inductor then there is no instantaneous effect on the flux associated with inductor. $\Rightarrow \phi$ = constant = $L_i I_i = L_f I_f$.

Q. **At t = 0 switch is closed (shown in figure). After a long time suddenly the inductance of the inductor is made $\frac{1}{\eta}$ times. Find out instantaneous current just after the operation**

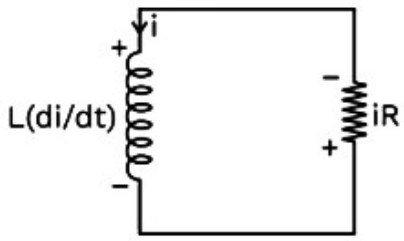


Sol. $L_i I_i = L_f I_f \Rightarrow L \times \frac{E}{R} = \frac{L}{\eta} I_f$

$I_f = \eta \frac{E}{R}$

Decay of Current in the Circuit Containing Resistor and Inductor

Let the initial current in the circuit be I_0. At any time t, let the current be i and let its rate of change at this instant be $\frac{di}{dt}$.



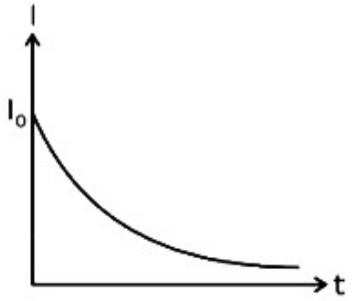
$$L \cdot \frac{di}{dt} + iR = 0, \quad \frac{di}{dt} = -\frac{iR}{L}$$

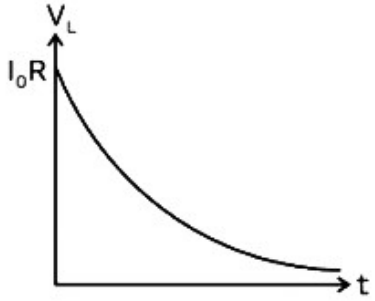
$$\int_{I_0}^{I} \frac{di}{i} = -\int_{0}^{t} \frac{R}{L} \cdot dt \Rightarrow \ln\left(\frac{i}{I_0}\right) = -\frac{iR}{L} \quad \text{or} \quad i = I_0 e^{-\frac{Rt}{L}}$$

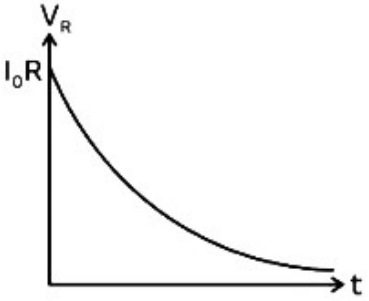
$$V_R = iR = i_0 R e^{-\frac{Rt}{L}}$$

$V_L = V_R$

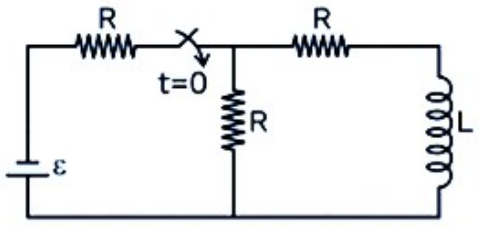
Current after one time constant $\left(\tau = \frac{L}{R}\right)$: I = I = 0.37 I_0 = 37% of initial current.



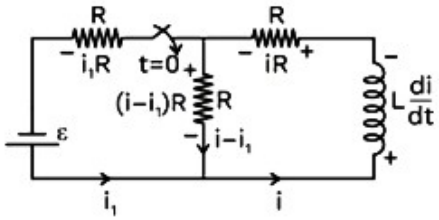




Q. **In the following circuit the switch is closed at t = 0. Initially there is no current in inductor. Find out current the inductor coil as a function of time.**



Sol. At any time t



$$-\varepsilon + i_1R - (i - i_1)R = 0$$

$$-\varepsilon + 2i_1R - iR = 0$$

$$i_1 = \frac{iR + \varepsilon}{2R}$$

Now, $-\varepsilon + i_1R + iR + L \cdot \frac{di}{dt} = 0$

$$-\varepsilon + \left(\frac{iR + \varepsilon}{2}\right) + iR + L \cdot \frac{di}{dt} = 0 \quad \Rightarrow \quad -\frac{\varepsilon}{2} + \frac{3iR}{2} = -L \cdot \frac{di}{dt}$$

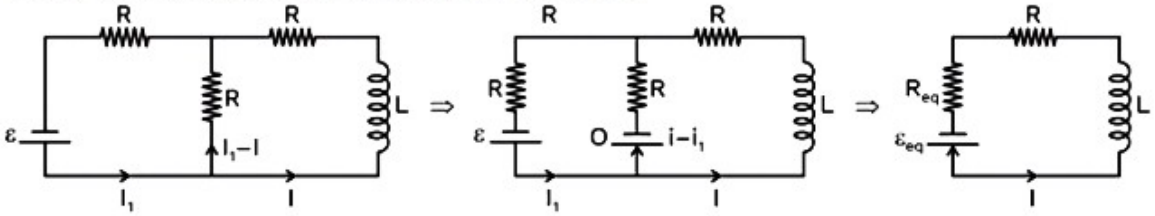
$$\left(\frac{-\varepsilon + 3iR}{2}\right)dt = -L \cdot di \quad \Rightarrow \quad -\frac{dt}{2L} = \frac{di}{-\varepsilon + 3iR}$$

$$-\int_0^t \frac{dt}{2L} = \int_0^i \frac{di}{-\varepsilon + 3iR} \quad \Rightarrow \quad -\frac{t}{2L} = \frac{1}{3R}\ln\left(\frac{-\varepsilon + 3iR}{-\varepsilon}\right)$$

$$-\ln\left(\frac{-\varepsilon + 3iR}{-\varepsilon}\right) = \frac{3Rt}{2L} \quad \Rightarrow \quad i = +\frac{\varepsilon}{3R}\left(1 - e^{-\frac{3Rt}{2L}}\right)$$

Other Method

Given circuit can be redrawn at any time t as



$$\varepsilon_{eq} = \frac{\frac{\varepsilon}{R} + \frac{0}{R}}{\frac{1}{R} + \frac{1}{R}} = \frac{\varepsilon}{2},\ R_{eq} = \frac{R}{2},\ \tau = \frac{L}{R + \frac{R}{2}} = \frac{2L}{3R}$$

Current through inductor, $I = \frac{\frac{\varepsilon}{2}}{\frac{3R}{2}}\left(1 - e^{-\frac{3Rt}{2L}}\right) = \frac{\varepsilon}{3R}\left(1 - e^{-\frac{3Rt}{2L}}\right)$

Writing KVL in left loop of first diagram

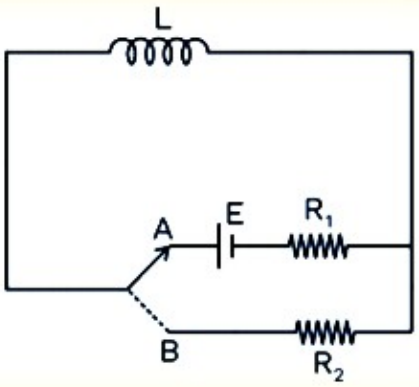
$$\Rightarrow \quad \varepsilon - (I_1 - I)R - I_1R = 0 \Rightarrow I_1 = \frac{\varepsilon + IR}{2R} = \frac{\varepsilon}{2R} + \frac{I}{2}$$

$$I = \frac{\varepsilon}{2R} + \frac{\varepsilon}{6R}\left(1 - e^{-\frac{3Rt}{2L}}\right)$$

Q. **In given circuit initially switch is at position A and circuit is in steady state. Now switch is shifted to position B. Taking this instant as t = 0, find**

(i) Current in inductor as function of time

(ii) Heat produced in R_2 from t = 0 to $t = \frac{L \ln 4}{R_2}$



Sol. (i) When switch is at position 'A', and circuit is in steady state steady state, steady state current $I_0 = \frac{E}{R_1}$

Now when switch is shifted to position B, let current in circuit at any time t is I. Writing KVL

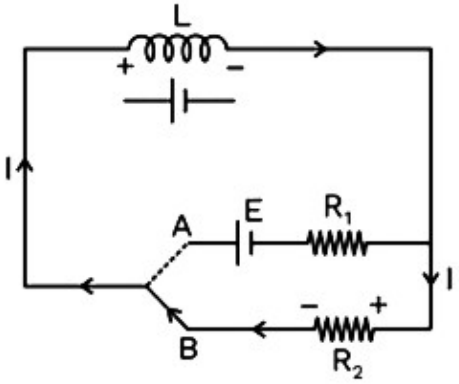
$0 - IR_2 - V_L = 0$

$$-\frac{V_L}{R_2} = I \Rightarrow -\frac{L}{R_2}\frac{dI}{dt} = I$$

$$\int_{I_0}^{I} \frac{dI}{I} = \int_0^t -\frac{R_2}{L}dt \Rightarrow \left[\ln I\right]_{I_0}^{I} = -\frac{R_2 t}{L}$$

$$\ln \frac{I}{I_0} = -\frac{R_2 t}{L} \Rightarrow I = I_0 \cdot e^{-\frac{R_2 t}{L}}$$

$$I = \frac{E}{R_1} e^{-\frac{R_2 t}{L}}$$



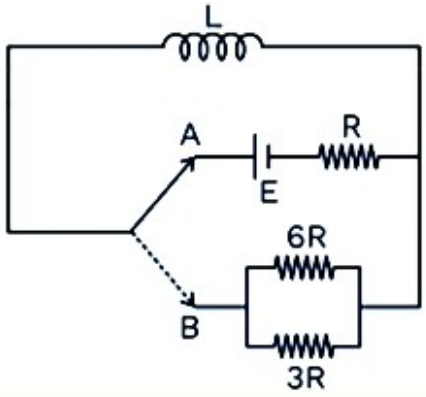
(ii) $\text{Heat} = \int I^2 R dt = \int_0^{T \ln 4} \frac{E^2}{R_1^2} e^{-\frac{2R_2 t}{L}} R_2 dt$

$$= \frac{E^2}{R_1^2} R_2 \left[\frac{e^{-\frac{2R_2 t}{L}}}{\frac{-2R_2}{L}}\right]_0^{T \ln 4} = \frac{1}{2}\frac{E^2 R_2}{R_1^2}\frac{L}{R_2}\left[1 - \frac{1}{16}\right]$$

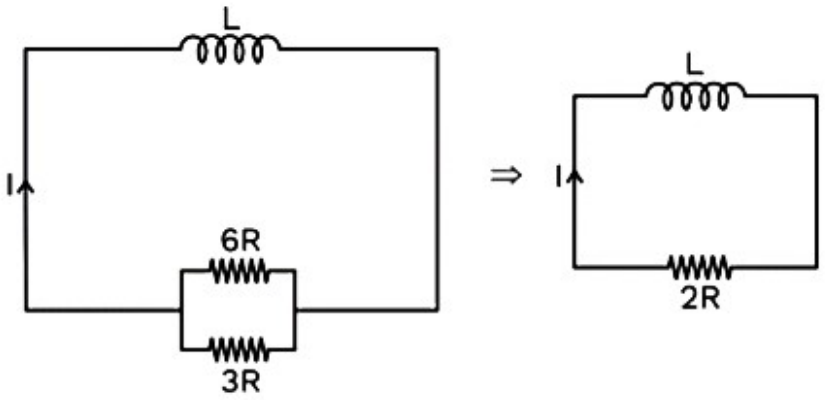
$$\text{Heat} = \frac{15E^2 L}{32R_1^2}$$

$$\text{or Heat} = \frac{1}{2}L\left(\frac{E}{R_1}\right)^2 - \frac{1}{2}L\left(\frac{E}{4R_1}\right)^2 = \frac{1}{2}L(I_i)^2 - \frac{1}{2}L(I_f)^2$$

Q. **In given circuit initially switch is at position 'A' and circuit has reached steady state. Then switch is shifted to position 'B' (take this time as t = 0). Find heat loss in 6R from t = 0 to $t = \frac{L\ln 2}{R}$.**



Sol. When switch is shifted to position B



Time constant of circuit, $\tau = \frac{L}{2R}$; at t = 0, $I = I_0 = \frac{e}{R}$

At $t = t$, $I = I_0 e^{-\frac{t}{\tau}} = \frac{E}{R} e^{-\frac{2R}{L}t}$, At $t = \frac{L\ln 2}{R}$, $I = \frac{E}{R} e^{-2\ln 2} = \frac{E}{4R}$

Total heat loss from t = 0 to $t = \frac{L\ln 2}{R}$, $H = \frac{1}{2}LI_0^2 - \frac{1}{2}LI^2 = \frac{1}{2}L\left(\frac{E^2}{R^2} - \frac{E^2}{16R^2}\right) = \frac{15E^2L}{32R^2}$

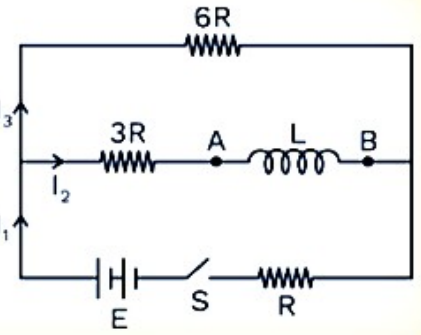
$P = \frac{V^2}{R}$, $\frac{P_{6R}}{P_{3R}} = \frac{1}{2} \Rightarrow \frac{H_{6R}}{H_{3R}} = \frac{1}{2}$

$H_{6R} = \frac{1}{3}$ (Total heat) $= \frac{1}{3}\left(\frac{15E^2L}{32R^2}\right) = \frac{5E^2L}{32R^2}$

Q. **In the given circuit switch 'S' is closed at t = 0.**

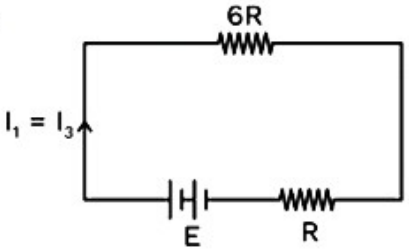
(i) Find currents I_1, I_2 and I_3 at t = 0 and t → ∞.

(ii) The circuit reaches to steady state and now suddenly the self inductance is made twice. Find current through all branches just after this moment and rate of change of current through inductor just after this moment.



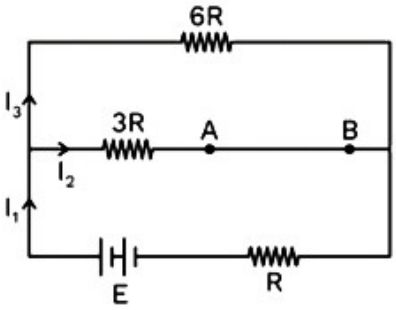
Sol. (i) At t = 0, current in branch containing inductor is zero ⇒ $I_2 = 0$

$$I_1 = I_3 = \frac{E}{7R}$$



At t → ∞ inductor behave as conducting wire.

$$I_1 = \frac{E}{3R}, \frac{I_2}{I_3} = \frac{6R}{3R} = \frac{2}{1} \Rightarrow I_2 = \frac{2E}{9R}, I_3 = \frac{E}{9R}$$



(ii) Let new currents be I_1', I_2' and, I_3'

ϕ through inductor will not change suddenly so,

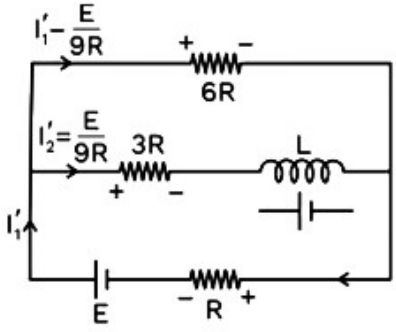
$$\phi = LI;\ L' = 2L \Rightarrow I_2^1 = \frac{E}{9R}$$

$$-I_1'R + E - \left(I_1' - \frac{E}{9R}\right)6R = 0 \Rightarrow -I_1'R + E - I_1'6R + \frac{2E}{3} = 0$$

$$\Rightarrow -7I_1'R = -\frac{5E}{3} \Rightarrow I_1 = \frac{5E}{21R}$$

$$2L\frac{dI_2'}{dt} + \frac{E}{9R}\cdot 3R - \left(I_1' - \frac{E}{9R}\right)6R = 0 \Rightarrow 2L\frac{dI_2'}{dt} = \frac{3E}{7}$$

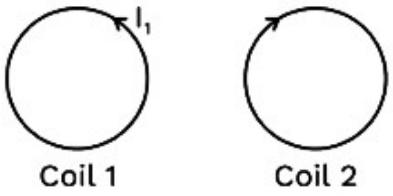
$$\frac{dI_2'}{dt} = \frac{3E}{14L}$$



Mutual Inductance

Mutual Inductance

Suppose there are two coils placed nearby. If current flowing through one coil changes with time, some emf is induced in the second coil. This phenomena is called mutual inductance



ϕ_{21} = magnetic flux linked with coil 2 due to current in coil 1

$$\phi_{21} \propto I_1$$

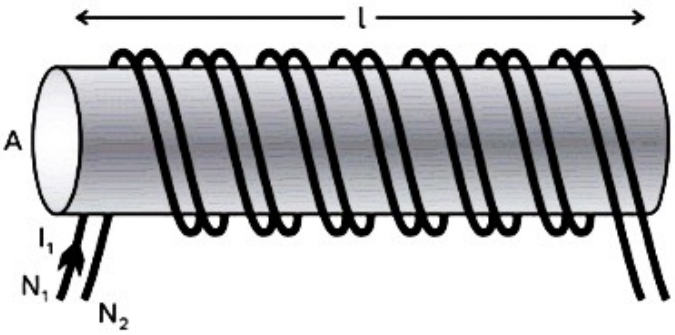
$$\boxed{\phi_{21} = M_{21} I_1} \; ; \; \boxed{M_{12} = M_{12}}$$

$$\boxed{\varepsilon_{21} = -M_{21} \frac{dI_1}{dt}}$$

$M_{12} = M_{21} = M$ = mutual inductance coefficient

Relation between Mutual Inductance and Self Inductance of two Coils

Two insulated wires are wound on the same hollow cylinder, so as to form two solenoids sharing a common air-filled core. Let l be the length of core, A the cross-sectional area of the core, N_1 the number of times the first wire is wound around the core, and N_2 the number of times the second wire is wound around the core. Find the mutual inductance of the two solenoids, neglecting the end effects.



$$B_1 = \mu_0 \frac{N_1}{l} I_1$$

$$\phi_{21} = N_2 B_1 A$$

$$\phi_{21} = N_2 \left(\frac{\mu_0 N_1 I_1}{l} \right) A \text{ and } \phi_{21} = M I_1$$

Comparing these two relations, we get

$$\boxed{M = \frac{\mu_0 N_1 N_2}{l} A}$$

$$\boxed{M = \sqrt{L_1 L_2}}$$

$$\left(\because \; L_1 = \frac{\mu_0 N_1^2 A}{l} \text{ and } L_2 = \frac{\mu_0 N_2^2 A}{l} \right)$$

In general $M \leq \sqrt{L_1 L_2}$

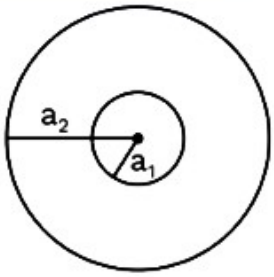
Q. **Find the mutual inductance of two concentric coils of radii a_1 and a_2 ($a_1 << a_2$) if**

(i) The planes of coils are same.

(ii) The planes of coil are perpendicular.

(iii) The planes of coils make θ angle with each other.

Sol. (i) Let a current i flow in coil of radius a_2



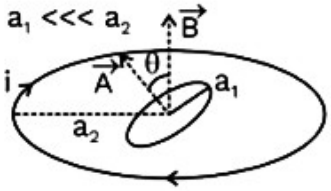
Magnetic field at the centre of coil $= \frac{\mu_0 i}{2a_2}$

Magnetic flux associated with coil of radius $a_1 = \frac{\mu_0 i}{2a_2}\pi a_1^2$

or $Mi = \frac{\mu_0 i}{2a_2}\pi a_1^2$ or $M = \frac{\mu_0 \pi a_1^2}{2a_2}$

(ii) Let a current i flow in the coil of radius a_2. The magnetic field at the centre of this coil will now be parallel to the plane of smaller coil and hence no flux will pass through smaller coil, hence M = 0.

(iii) If i current flows in the larger coil, magnetic field produced at the centre will be perpendicular to the plane of larger coil.



Now the area vector of smaller coil which is perpendicular to the plane of smaller coil will make an angle θ with the magnetic field

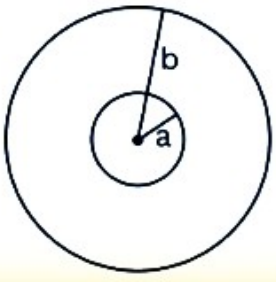
Thus flux $= \vec{B}.\vec{A} = \frac{\mu_0 i}{2a_2}.\pi a_1^2.\cos\theta$ or $M = \frac{\mu_0 \pi a_1^2 \cos\theta}{2a_2}$

Q. **Figure shows two concentric coplanar coils with radii a and b (a << b). A current i = 2t flows in the smaller loop. Neglecting self inductance of larger loop**

(a) Find the mutual inductance of the two coils

(b) Find the emf induced in the larger coil

(c) If the resistance of the larger loop is R find the current in it as a function of time



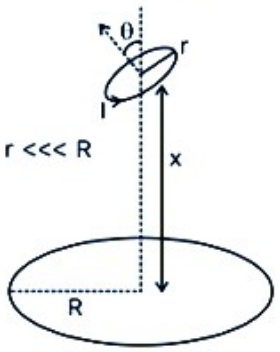
Sol. (a) To find mutual inductance, it does not matter in which coil we consider current and in which coil flux is calculated (Reciprocity theorem). Let current i be flowing in the larger coil. Magnetic field at the centre $= \frac{\mu_0 i}{2b}$.

Flux through the smaller coil $= \frac{\mu_0 i}{2b}\pi a^2 \quad \therefore \quad M = \frac{\mu_0}{2b}\pi a^2$

(b) $\left|\text{emf induced in larger coil}\right| = M\left[\left(\frac{di}{dt}\right)\text{ in smaller coil}\right] = \frac{\mu_0}{2b}\pi a^2\ (2) = \frac{\mu_0 \pi a^2}{b}$

(c) Current in the larger coil $= \frac{\mu_0 \pi a^2}{bR}$

Q. **Two coils are located as shown in diagram. A current I flows through the smaller coil of radius r. Find the flux through coil of radius 'R'**



Sol. To calculate mutual inductance between the coils, take current I_1 through coil of radius R and write flux associated with smaller coil of radius r.

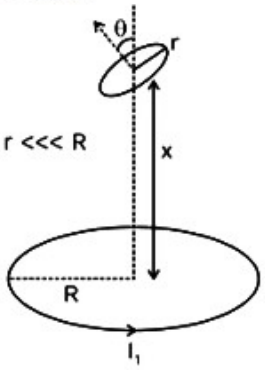
As $M_{21} = M_{12}$ so inverse can be done

ϕ (flux through smaller coil)

$$= \frac{\mu_0 I_1 R^2}{2(R^2 + x^2)^{3/2}}\pi r^2 \cos\theta = \left(\frac{\mu_0 R^2 \pi r^2 \cos\theta}{2(R^2 + x^2)^{3/2}}\right) I_1$$

$$M = \frac{\mu_0 R^2 \pi r^2 \cos\theta}{2(R^2 + x^2)^{3/2}}$$

ϕ (flux through bigger coil) = mutual inductance × current through smaller coil MI



Mutual Inductance Depends on

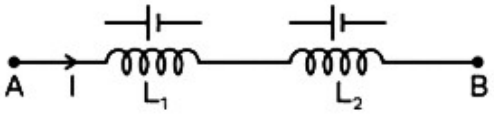
(i) Shape and size of coil
(ii) Relative separation between them
(iii) Medium

Combination of inductors

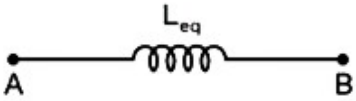
Combination of inductors

Series

$$V_A - V_B = L_1\frac{dI}{dt} + L_2\frac{dI}{dt} = L_{eq}\frac{dI}{dt}$$



$\boxed{L_{eq} = L_1 + L_2}$ ⇒ Neglecting mutual inductance

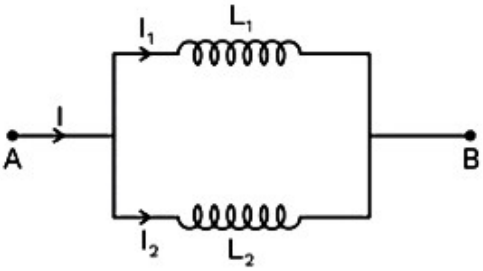


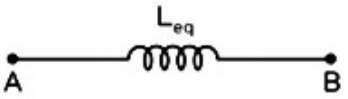
$\boxed{L_{eq} = L_1 + L_2 \pm 2M}$ ⇒ considering mutual inductance

Parallel

$I = I_1 + I_2$

$$\frac{dI}{dt} = \frac{dI_1}{dt} + \frac{dI_2}{dt}$$





$$\frac{V_A - V_B}{L_{eq}} = \frac{V_A - V_B}{L_1} + \frac{V_A - V_B}{L_2} \Rightarrow \frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}$$

Generator

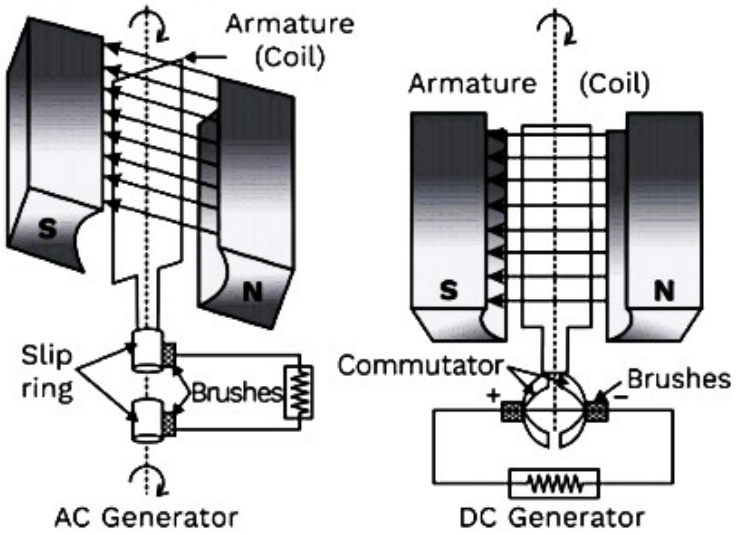
Generator or Dynamo

Generator or dynamo is an electrical device which converts mechanical energy into electrical energy. Working of generators is based on the principle of electromagnetic induction.

Generators are of two types:

(a) A.C. generator : If the current produced by the generator is alternating, then the generator is called A.C. generator.

(b) D.C. generator : If the current produced by the generator is direct current, then the generator is called D.C. generator.



Generator consists of the following parts.

(a) Armature (coil) (b) Magnet

(c) Slip rings (d) Brushes

In D.C. generator commutator is used in place of slip rings.

In order to produce the magnetic field in big generators several magnetic poles are used. In these generators the armature coils are kept stationary and magnetic pole pieces are made to rotate around the armature. The frequency of alternating current produced by generator of multi poles is

$$= \frac{\text{number of poles} \times \text{rotational frequency}}{2} = \frac{Nn}{2}$$

Energy loss in generators : The loss of energy is due to the following reasons :

(a) Flux leakage (b) Copper losses

(c) Eddy current losses (d) Hysteresis losses

(e) Mechanical losses

Efficiency of generator : Practical efficiency of a generator

$$= \frac{\text{Electrical power generated by the generator}}{\text{Mechanical energy given to the generator}}$$

Practical efficiencies of big generators are about 92% to 95%.